Sergio Lozano

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Summary

Computer Science M.S. graduate building Python backends, AI applications, and data pipelines. Project experience spans transformer training in JAX, benchmarked Rust inference, biomedical retrieval and fine-tuning, and REST APIs. Biology background supports scientific problem-solving and research documentation.

Technical Skills

Languages: Python, Rust, SQL, C++, JavaScript
AI and ML: PyTorch, JAX, scikit-learn, Hugging Face, QLoRA, LangGraph, RAG, ChromaDB, SHAP
Backend and Data: FastAPI, Flask, REST APIs, SQLite, pandas, NumPy, Streamlit
Tools: Git, Docker, Linux, Weights & Biases | **Language:** Fluent Spanish

Selected Projects

Forge — Transformer Training and Rust Inference JAX, Rust, Python

- Built a 13.69M-parameter decoder-only transformer from scratch in JAX with custom attention, RoPE, and RMSNorm; trained on TinyStories for 20,000 steps.
- Reimplemented inference in Rust and verified numerical agreement with JAX; benchmarked 105.2 tokens/sec versus 13.7 for the Python baseline (7.7× faster) on project hardware.
- Benchmarked KV caching, batching, and INT8 quantization; reduced memory by 58% with a measured 0.01% perplexity increase using INT8.

BioLyra — Biomedical AI Assistant PyTorch, FastAPI, LangGraph, ChromaDB

- Fine-tuned Phi-3.5-mini on PubMedQA using 4-bit QLoRA; indexed approximately 2,500 PubMed abstract chunks with BioBERT embeddings in ChromaDB.
- Routed queries between conversation, document retrieval, and live NCBI Entrez search using LangGraph; exposed the assistant through a FastAPI endpoint and Gradio interface.
- Compared base and fine-tuned outputs using BERTScore and tracked experiments in Weights & Biases.

Job Application Pipeline Python, SQLite, Claude API

- Built a pipeline ingesting Greenhouse, Lever, and USAJobs postings into SQLite with four-level deduplication and explainable scoring against multiple résumé profiles.
- Integrated the Claude API for résumé and cover-letter tailoring, with qualification-preservation checks and explicit confirmation before application submission.

Stock & ETF Analyzer Python, FastAPI

- Built a research dashboard for U.S. stocks and ETFs with interactive price and volume charts, technical indicators, and watchlists; implemented a FastAPI backend with reproducible data snapshots and a prediction journal to evaluate forecasts against observed outcomes.

Education

University of Texas Rio Grande Valley Edinburg, TX
M.S. in Computer Science | GPA: 3.80/4.00 May 2026
B.S. in Biology, Minor in Chemistry 2023

Experience

Biochemistry Lab Assistant — UTRGV 2022–2023

- Executed DNA extraction and PCR workflows; maintained detailed laboratory records and research documentation.

Professional Development

AI Engineer Core Track — LLM Engineering, RAG, QLoRA, Agents; Udemy June 2026
AI Engineer Agentic Track — The Complete Agent & MCP Course; Udemy July 2026